Dr Filippo Pecci

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Research Experience

Research Associate, Computational Optimisation — **Imperial College London**
Department of Civil and Environmental Engineering — *April 2018–present*

- Fundamental research into tailored optimisation algorithms to solve large scale single and multi-objective mixed-integer non-linear problems for optimal design and operation of water systems;
- Development of optimisation tools for optimal network design, control, model identification, and fault estimation in water systems;

Qualifications

PhD in Computational Optimisation — **Imperial College London**
Department of Civil and Environmental Engineering — *2014–2018*
Thesis: Optimal design-for-control of water supply networks by mixed integer programming

- Modelling and analysis of large scale water distribution networks;
- Novel tailored global optimisation methods to solve mixed integer nonlinear programs for optimal design-for-control of water distribution networks;
- Application of mathematical optimisation methods to multi-objective optimal design-for-control problems in water distribution networks;

MSc in Mathematics — **Universita' degli studi di Padova**
Department of Mathematics — *2011–2014*
Emphasis on PDEs, stochastic analysis, applied linear algebra, computational methods.
Thesis: Hierarchical Stratifications of Pareto Sets

BSc in Mathematics — **Universita' degli studi di Padova**
Department of Mathematics — *2008–2011*

Teaching Experience

Lecturing and tutorials — **Imperial College London**
Department of Civil and Environmental Engineering — *2014-present*

- Lectures for BSc modules on convex optimisation with applications in optimal reservoir control, and parameter estimation in water distribution networks.
- Tutorials on applications of convex optimisation to problems in image in-painting, optimal control, machine learning, and hydraulic model calibration.
- Lectures and tutorials for MSc modules on modelling and optimisation of water distribution networks.
- Small-group tutorials on mathematics for MSc students, with focus on linear algebra and calculus;
- Graduate Teaching Assistant for MSc modules in Hydroinformatics and Water distribution and Distribution.

Supervision — **Imperial College London**
Department of Civil and Environmental Engineering — *2014-present*

- Technical supervision of 5 PhD students working on optimal design, model identification and fault estimation problems for water distribution networks.
- Co-supervision of 4 final year MSc projects on analysis and optimisation of water distribution networks.

Publications

Peer-reviewed journal articles

[J15] F. Pecci, I. Stoianov, and A. Ostfeld. "Convex Heuristics for Optimal Placement and Operation of Valves and Chlorine Boosters in Water Networks". In: *Journal of Water Resources Planning and Management* 148.2 (2022), pp. 1–14. DOI: 10.1061/(ASCE)WR.1943-5452.0001509.

[J14] A.-J. Ulusoy, H. A. Mahmoud, F. Pecci, E. C. Keedwell, and I. Stoianov. "Bi-objective design-for-control for improving the pressure management and resilience of water distribution networks". In: *Water Research* (2022), p. 118914. DOI: https://doi.org/10.1016/j.watres.2022.118914. URL: https://www.sciencedirect.com/science/article/pii/S0043135422008612.

[J13] C. Blocher, F. Pecci, and I. Stoianov. "Prior Assumptions for Leak Localisation in Water Distribution Networks with Uncertainties". In: *Water Resources and Management* (2021). DOI: 10.1007/s11269-021-02988-z.

[J12] F. Pecci, I. Stoianov, and A. Ostfeld. "Relax-tighten-round algorithm for optimal placement and control of valves and chlorine boosters in water networks". In: *European Journal of Operational Research* 295.2 (2021), pp. 690–698. DOI: 10.1016/j.ejor.2021.03.004.

[J11] A.-J. Ulusoy, F. Pecci, and I. Stoianov. "Bi-objective design-for-control of water distribution networks with global bounds". In: *Optimization and Engineering* (2021). Published online. DOI: 10.1007/s11081-021-09598-z.

[J10] C. Blocher, F. Pecci, and I. Stoianov. "Localizing Leakage Hotspots in Water Distribution Networks via the Regularization of an Inverse Problem". In: *Journal of Hydraulic Engineering* 146.4 (2020). DOI: 10.1061/(ASCE)HY.1943-7900.0001721.

[J9] D. Nerantzis, F. Pecci, and I. Stoianov. "Optimal control of water distribution networks without storage". In: *European Journal of Operational Research* 284.1 (2020), pp. 345–354. DOI: 10.1016/j.ejor.2019.12.011.

[J8] F. Pecci, P. Parpas, and I. Stoianov. "Sequential Convex Optimization for Detecting and Locating Blockages in Water Distribution Networks". In: *Journal of Water Resources Planning and Management* 146.8 (2020). DOI: 10.1061/(ASCE)WR.1943-5452.0001233.

[J7] A.-J. Ulusoy, F. Pecci, and I. Stoianov. "An MINLP-Based Approach for the Design-for-Control of Resilient Water Supply Systems". In: *IEEE Systems Journal* 14.3 (2020), pp. 4579–4590. DOI: 10.1109/JSYST.2019.2961104.

[J6] A. Waldron, F. Pecci, and I. Stoianov. "Regularization of an Inverse Problem for Parameter Estimation in Water Distribution Networks". In: *Journal of Water Resources Planning and Management* 146.9 (2020). DOI: 10.1061/(ASCE)WR.1943-5452.0001273.

[J5] F. Pecci, E. Abraham, and I. Stoianov. "Global optimality bounds for the placement of control valves in water supply networks". In: *Optimization and Engineering* 20.2 (2019), pp. 457–495. DOI: 10.1007/s11081-018-9412-7.

[J4] F. Pecci, E. Abraham, and I. Stoianov. "Model Reduction and Outer Approximation for Optimizing the Placement of Control Valves in Complex Water Networks". In: *Journal of Water Resources Planning and Management* 145.5 (2019). DOI: 10.1061/(ASCE)WR.1943-5452.0001055.

[J3] F. Pecci, E. Abraham, and I. Stoianov. "Penalty and relaxation methods for the optimal placement and operation of control valves in water supply networks". In: *Computational Optimization and Applications* 67.1 (2017), pp. 201–223. DOI: 10.1007/s10589-016-9888-z.

[J2] F. Pecci, E. Abraham, and I. Stoianov. "Quadratic head loss approximations for optimisation problems in water supply networks". In: *Journal of Hydroinformatics* 19.4 (2017), pp. 493–506. DOI: 10.2166/hydro.2017.080.

[J1] F. Pecci, E. Abraham, and I. Stoianov. "Scalable Pareto set generation for multiobjective co-design problems in water distribution networks: a continuous relaxation approach". In: *Structural and Multidisciplinary Optimization* 55.3 (2017), pp. 857–869. DOI: 10.1007/s00158-016-1537-8.

Peer-reviewed conference proceedings

[C4] F. Pecci, I. Stoianov, and A. Ostfeld. "Optimal Design-for-Control of Chlorine Booster Systems in Water Networks via Convex Optimization". In: *Proceedings of the European Control Conference 2022*. 2022.

[C3] F. Pecci, E. Abraham, and I. Stoianov. "Outer approximation methods for the solution of co-design optimisation problems in water distribution networks". In: *IFAC-PapersOnLine*. Vol. 50. 1. 2017, pp. 5373–5379. DOI: 10.1016/j.ifacol.2017.08.1069.

[C2] F. Pecci and I. Stoianov. "Optimising valve placement and pressure control for multi-feed sectors in water supply networks using outer approximation". In: Figshare, 2017. DOI: 10.15131/shef.data.5364196.v1. CCWI 2017 - 15th International Conference on Computing and Control for the Water Industry.

[C1] F. Pecci, E. Abraham, and I. Stoianov. "Mathematical programming methods for pressure management in water distribution systems". In: *Procedia Engineering*. Vol. 119. 1. 2015, pp. 937–946. DOI: 10.1016/j.proeng.2015.08.974. Computing and Control for the Water Industry (CCWI2015).

In preparation and submitted for publication

[S2] F. Pecci and I. Stoianov. "Bounds and Convex Heuristics for Multiobjective Sensor Placement in Water Networks". 2022. Preprint arXiv:2207.04897, submitted to Computers and Operations Research.

[S1] A. Waldron, A.-J. Ulusoy, F. Pecci, and I. Stoianov. "Principal Component Based Sampling for the Continuous Maintenance of Hydraulic Models". 2022. Accepted for publication in Water Research.

Patents

[B2] A. Waldron, F. Pecci, and I. Stoianov. "Online maintenance of hydraulic models for WSN through continuous monitoring and adaptive control". 2021. Filed. GB application number 2112111.6.

[B1] I. Stoianov, E. Abraham, and F. Pecci. "Management of liquid conduit systems". 2015. Granted. PCT/GB2016/054026. GB2545899B (2018), US11078650B2 (2021), EP3394697B1 (2021).

Reviews

[R1] A. Waldron, F. Pecci, and I. Stoianov. *Closure to "Regularization of an Inverse Problem for Parameter Estimation in Water Distribution Systems" by Alexander Waldron, Filippo Pecci, and Ivan Stoianov*. In: *Journal of Water Resources Planning and Management* vol. 147.12 (2021). DOI: 10.1061/(ASCE)WR.1943-5452.0001490.

Others

[A2] F. Pecci. *Optimal design for control of water supply networks by mixed integer programming*. 2018. PhD Thesis. Imperial College London.

[A1] A. Lovison and F. Pecci. *Hierarchical stratification of Pareto sets*. arXiv:1407.1755. 2014. MSc Dissertation.

Conference presentations

1. F. Pecci, I. Stoianov, and A. Ostfeld. Optimal Design-for-Control of Chlorine Booster Systems in Water Networks via Convex Optimization. European Control Conference 2022, London, United Kingdom, 12-15 July, 2022.
2. I. Stoianov, F. Pecci. Mathematical optimisation for intelligent water distribution networks: model calibration, and event detection and localisation. Invited seminar part of the series Control & Optimisation Seminars, Imperial College London, 22 Gennaio 2020.
3. F. Pecci, E. Abraham, I. Stoianov. Tight Convex Relaxations for Optimal Design and Control Problems in Water distribution Networks. 17th Computing and Control for the Water Industry (CCWI), Exeter, United Kingdom, 1-4 September, 2019.
4. F. Pecci, P. Parpas, and I. Stoianov. Non-linear inverse problems via sequential convex optimization. Invited presentation at 6th International Conference on Continuous Optimization (ICCOPT), Berlin, Germany, 3 - 8 August, 2019.
5. F. Pecci, E. Abraham, I. Stoianov. A branch and bound method for globally optimising valve locations in water distribution networks. 6th International Conference on Engineering Optimization (EngOpt), Lisbon, Portugal, 17 - 19 September, 2018.
6. F. Pecci, E. Abraham, I. Stoianov. Outer approximation methods for the solution of co-design optimisation problems in water distribution networks. Invited presentation at 20th IFAC World Congress, Toulouse, France, 9 - 14 July, 2017.
7. F. Pecci, E. Abraham, I. Stoianov. Multiobjective pressure optimization in water distribution systems. 14th Computing and Control for the Water Industry (CCWI), Amsterdam, the Netherlands, 7-9 November, 2016 (Poster Presentation).
8. F. Pecci, E. Abraham, I. Stoianov. Mathematical programming methods for pressure management in water distribution systems. 13th Computing and Control for the Water Industry (CCWI), Leicester, United Kingdom, 2-4 September, 2015.